Chapter 1

Lecture 1

ECEN-743 Dr. Dileep Kalathil MW 4:10 PM - 5:25 PM BLOC 113

1.1 Introduction – 1/12

We define machine learning as a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other... - Murphy

Supervised learning maps an input to an output based on example input-output pairs (labelled data)

Examples: image classification, email spam filtering, speech recognition

Unsupervised learning is the task of inferring a function that describes the structure of "unlabeled" data

Examples: clustering, dimensionality reduction (e.g. PCA, autoencoder)

Reinforcement learning is an area of ML concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward

Reinforcement learning challenges: no prior data (RL agent generates data, by taking action in an unknown environment), data generated is not i.i.d, no supervisor (optimal policy has to be learnt from the rewards observed in the self-generated data), actions have long term consequences (rewards are often delayed), dynamical systems issues (feedback and stability)

AlphaGo

1.2 Lecture 4 - Jan 28

1.2.1 Q-value function

Value function of policy π is defined as

$$V_\pi(s) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s]$$

Action-value function for policy π is defined as

$$Q_\pi(s, a) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s, a_0 = a]$$

It is easy to show

$$Q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s'|s, a) V_\pi(s')$$

this is the instantaneous reward for (s, a) and the expected value of the next state.

1.3 Value Policy Iteration

Fixed point

Let $f : \mathcal{U} \to \mathcal{U}$. u^* is a fixed point of $f(\cdot)$ if $f(u^*) = u^*$.

Let $f : \mathcal{U} \to \mathcal{U}$ and let $\|\cdot\|$ be a norm defined on $\mathcal{U}$. We say $f(\cdot)$ is a contraction mapping (or simply, contraction) if $\|f(u) - f(v)\| \leq \alpha \|u - v\|$ for an $\alpha \in (0, 1)$ and for all $u, v \in \mathcal{U}$.

Banach Fixed Point Theorem.

$$\|u_{n+1} - u_n\| \leq \alpha^n \|u_1 - u_0\|$$

converges when $\alpha \leq 1$. We can show that the sequence $\{u_0, u_1, u_2, \dots\}$ is a Cauchy sequence. Since $\mathcal{U}$ is closed and bounded, the sequence converges, i.e., there exists a limit $u^* = \lim_{k\to\infty} u_k$. Since $f()$ is a contraction, it is also a continuous function. Then, $u^* = \lim_{k\to\infty} u_k = f(\lim_{k\to\infty} u_k) = f(u^*)$.

Proof of uniqueness: suppose there exists another fixed point $\bar{u}^* \neq u^*$. Then,

$$\|u^* - \bar{u}^*\| = \|f(u^*) - f(\bar{u}^*)\| \leq \alpha \|u^* - \bar{u}^*\|.$$

This is a contradiction, so $u^* = \bar{u}^*$.

1.4 Bellman Optimality Equation

Let V^* be the optimal value function. Then, V^* satisfies the equation

$$V^*(s) = \max_a \left(r(s,a) + \gamma \sum_{s'} P(s'|s,a)V^*(s') \right)$$

Suppose the current state is s and we take action a. From the next state onwards, we follow the optimal policy π^*. What is the 'value' of state s (expected cumulative discounted reward) under this procedure?

$$V(s) = r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V^*(s')]$$

reward for (s, a) plus *gamma* times the expected value of the next state under the optimal policy π^*. The best action we can take in the current state s is the action which maximizes the value $V(s)$. The 'value' of state s if we take this action is

$$\max_a (r(s,a) + \gamma \mathbb{E}_{s \sim P(\cdot|s,a)}[V^*(s')])$$

1.5 Value Iteration

Bellman Operator $T : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ is defined as

$$(TV)(s) = \max_a (r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V(s')])$$

Aim to design feedback, with $V_{k+1} = TV_k$, for which we want to prove converge.

Proposition: Bellman operator T is a contraction with respect to $\|\cdot\|_\infty$, i.e., for any $V_1, V_2 \in \mathcal{R}^{|S|}$, $\|TV_1 - TV_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$.

A useful result for the proof. Let g, h be any two real-valued function. Then,

$$|\max_a g(x,a) - \max_a h(x,a)| \leq \max_a |g(x,a) - h(x,a)|$$

Proof (of contraction property)

$$\begin{aligned}
|(TV_1)(s) - (TV_2)(s)| &= |\max_{a \in \mathcal{A}} (r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V_1(s')]) - \max_{a \in \mathcal{A}} (r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V_2(s')])| \\
&\leq \max_{a \in \mathcal{A}} |\gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V(s')]) - ||\gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V(s')]) \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V_2(s')])| \\
&\leq \max_{a \in \mathcal{A}} \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[|V_1(s') - V_2(s')|] \\
&\leq \max_{a \in \mathcal{A}} \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[\|V_1 - V_2\|_\infty] = \gamma \|V_1 - V_2\|_\infty
\end{aligned}$$

this implies

$$\|TV_1 - TV_2\|_\infty \leq \gamma ...$$

Chapter 2

Lectures 6-10

2.1 February 2, 2026

Bellman Optimality Equation

$$V^*(s) = \max_a \left(r(s,a) + \gamma \sum_{s'} P(s'|s,a) V^*(s') \right)$$

The statement that this is a fixed-point has nothing to do with contraction. We want to perform an iterative loop using the Bellman operator. The optimal value function V^* is the unique fixed point of the Bellman operator T, i.e., $TV^* = V^*$. Value iteration, defined as $V_{k+1} = TV_k$, with an arbitrary initial value V_0 will converge to the optimal value function V^*, i.e. $\lim_{k\to\infty} V_k = V^*$.

Optimal policy π^* can be computed as

$$\pi^*(s) = \arg\max_a (r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V^*(s')])$$

Optimal Q-value function of a policy π

$$Q_\pi(s,a) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s, a_0 = 0]$$

Just like optimal value function, you can define an optimal Q-value function

$$Q^*(s,a) = \max_\pi Q_\pi(s,a)$$

There exists a conversion from V^* to Q^* through

$$Q^*(s,a) = r(s,a) = \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[V^*(s')]$$

And from Q^* to V^*

$$V^*(s) = \max_a Q^*(s,a)$$

Also from π^* to Q^*

$$\pi^*(s) = \arg\max_a Q^*(s,a)$$

In some problems, you may not know the transition probability.

Optimal Q-value function Q^* satisfies the Bellman equation

$$Q^*(s,a) = r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[\max_b Q^*(s',b)]$$

Define the mapping $F : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \to \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ as

$$(FQ)(s,a) = r(s,a) = \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)}[\max_b Q(s',b)]$$

2.1.1 Policy Iteration